PySpark Setup on Windows with VS Code

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Follow these 8 steps from top to bottom and you will run your first PySpark program, using only VS Code and its built-in terminal.

Before you begin

You will install four things. Every command below is typed in the VS Code terminal, so you never need a separate black window.

What	Why you need it	Version
VS Code	The editor where you type code and commands	Latest
Python	The language PySpark is written for	3.11
Java (JDK)	The engine that runs Spark underneath	17 or newer
PySpark	The Python tool for Spark	4.3.0

You do not need to install Apache Spark separately. PySpark brings what it needs.

Three words used below:

- **Terminal**: a text box inside VS Code where you type commands and press **Enter**.
- **Command**: one line of text you copy into the terminal. Run one command at a time.
- **Virtual environment**: a private box for this project's Python tools, so nothing clashes with other projects.

> **Tip:** To copy a command, click the copy button on the code box (or select it and press **Ctrl + C**). To paste in the terminal, press **Ctrl + V** or right-click.

Step 1: Install VS Code and open the terminal

1.1 Install VS Code

1. Open the VS Code download page.

2. Click **Windows** to download the installer.
3. Open the installer and accept the agreement.
4. On the **Select Additional Tasks** screen, tick **Add to PATH** and **Add "Open with Code" action to Windows Explorer file context menu**.
5. Click **Install**, then **Finish**.

Already have VS Code? Skip to 1.2.

1.2 Open the terminal inside VS Code

1. Open VS Code from the Start menu.
2. Click **Terminal** in the top menu, then **New Terminal**. (Shortcut: **Ctrl + `**, the key above Tab.)
3. A panel opens at the bottom of the window. This is your terminal.

1.3 Switch the terminal to Command Prompt

VS Code usually opens PowerShell. The commands in this guide are written for **Command Prompt (CMD)**, so switch once:

1. In the terminal panel, click the small **down arrow** (⌄) next to the **+** button.
2. Click **Command Prompt**.
3. The panel now shows a prompt ending in `>`, for example `C:\Users\YourName>`.

`YourName` is a placeholder. Windows shows your real user name.

Test that it works:

```
echo Hello
```

You should see `Hello` printed on the next line.

Step 2: Install Python 3.11

You will install Python by typing one command in the VS Code terminal.

2.1 Install Python

Run this command and wait until it finishes:

```
winget install -e --id Python.Python.3.11
```

If Windows asks for permission, click **Yes**. If it asks you to accept terms, type `Y` and press **Enter**.

> **No `winget`?** Use the manual way instead: open the Python for Windows page, download **Python 3.11.x (64-bit)**, run it, tick **Add python.exe to PATH**, and click **Install Now**.

2.2 Restart VS Code

Close VS Code completely and open it again. This lets the terminal find the new Python. Then open the terminal as in Step 1.2 and 1.3.

2.3 Check Python

```
py -3.11 --version
```

You should see something like this (the last number may differ):

```
Python 3.11.9
```

Now check pip, Python's tool installer:

```
py -3.11 -m pip --version
```

If both commands print a version, Python is ready. If you installed a different Python version, use that number instead of `3.11` everywhere in this guide.

2.4 Add the Python extension to VS Code

1. Click the **Extensions** icon on the left bar (four squares). Shortcut: **Ctrl + Shift + X**.
2. Type `Python` in the search box.
3. Find the one by **Microsoft** and click **Install**.

Step 3: Install Java 17

PySpark needs Java to run the Spark engine underneath.

3.1 Install Java

Run this command in the VS Code terminal:

```
winget install -e --id EclipseAdoptium.Temurin.17.JDK
```

> **No `winget`?** Open the Eclipse Temurin downloads page, choose **Windows**, **x64**, **JDK**, version **17** or newer, then download and run the installer.

3.2 Restart VS Code and check Java

Close VS Code, open it again, and open a Command Prompt terminal (Step 1.2 and 1.3). Then run:

```
java -version
```

The output must mention version `17` or newer.

Next, check the `JAVA_HOME` setting:

```
echo %JAVA_HOME%
```

- It prints a folder such as `C:\Program Files\Eclipse Adoptium\jdk-17...`: good, go to Step 4.
- It prints `%JAVA_HOME%` unchanged, or a wrong folder: do 3.3.

3.3 Set `JAVA_HOME` (only if it was missing or wrong)

1. In the terminal, list the installed Java folders:

   ```
   dir "C:\Program Files\Eclipse Adoptium"
   ```

2. Write down the folder name you see, for example `jdk-17.0.13.11-hotspot`. Your numbers will differ.
3. Set the variable (replace the folder name with yours):

   ```
   setx JAVA_HOME "C:\Program Files\Eclipse Adoptium\jdk-17.0.13.11-hotspot"
   ```

4. Close VS Code, open it again, open a Command Prompt terminal, and run:

   ```
   echo %JAVA_HOME%
   java -version
   where java
   ```

`JAVA_HOME` must point to the JDK **main folder**. Do not add `\bin` or `\java.exe` at the end.

> **Why restart?** A terminal reads settings only when it starts. An old terminal will not see the new Java.

Step 4: Make the project folder and open it in VS Code

4.1 Create the folder

In the terminal, run these two commands, one at a time:

```
mkdir "%USERPROFILE%\pyspark-basic"
```

```
cd /d "%USERPROFILE%\pyspark-basic"
```

The first makes a folder called `pyspark-basic` in your user folder. The second moves the terminal into it.

To see the full path, run `cd`. It should end with `pyspark-basic`, for example `C:\Users\YourName\pyspark-basic`.

4.2 Open the folder in VS Code

Run:

```
code .
```

The dot means "this folder". A new VS Code window opens with the **Explorer** panel on the left, showing `pyspark-basic` (empty for now). If VS Code asks **Do you trust the authors of the files in this folder?**, click **Yes, I trust the authors**.

In the new window, open the terminal again with **Terminal → New Terminal**. Pick **Command Prompt** from the ⌄ menu if it shows PowerShell. The prompt should now end with `pyspark-basic>`.

> **Already have a project folder?** Open it with **File → Open Folder**, then continue from Step 5. Do not make a second folder.

Step 5: Create and switch on the virtual environment

A virtual environment is a private box for this project's Python tools.

5.1 Create it

Make sure the terminal prompt ends with `pyspark-basic>`, then run:

```
py -3.11 -m venv .venv
```

A new folder named `.venv` appears in the VS Code Explorer. Do not edit anything inside it.

5.2 Switch it on (activate)

```
.venv\Scripts\activate.bat
```

Success looks like this, with `(.venv)` at the start of the line:

```
(.venv) C:\Users\YourName\pyspark-basic>
```

No `(.venv)`? Stop here. Check that the prompt ends with `pyspark-basic>` and run the activate command again.

5.3 Upgrade pip

```
python -m pip install --upgrade pip
```

5.4 Tell VS Code to use this environment

1. Press **Ctrl + Shift + P** to open the Command Palette.
2. Type `Python: Select Interpreter` and press **Enter**.
3. Choose the entry that shows `.venv` (for example `Python 3.11.x ('.venv')`).

The bottom-right corner of VS Code now shows your Python version with `.venv`.

5.5 Check you are using the right Python

```
where python
python --version
```

The first path printed must be inside `...\pyspark-basic\.venv\Scripts\`.

Step 6: Install PySpark

Check that `(.venv)` is still at the start of the terminal line. Then run:

```
python -m pip install pyspark==4.3.0
```

Wait until it finishes. It can take a few minutes, depending on your internet.

Check the version:

```
python -c "import pyspark; print(pyspark.__version__)"
```

Expected output:

```
4.3.0
```

Optional: see the package details.

```
python -m pip show pyspark
```

Py4J, the bridge between Python and Java, installs by itself. Official guide: [PySpark installation].

Step 7: Create your files in VS Code

You will make three files. VS Code saves them with the exact name you type, so there is no hidden `.txt` problem.

7.1 Create `first_spark.py`

1. In the **Explorer** panel on the left, hover over the `PYSPARK-BASIC` title bar.
2. Click the **New File** icon (a page with a plus sign).
3. Type `first_spark.py` and press **Enter**.
4. The empty file opens in the editor. Paste this code into it:

```
from pyspark.sql import SparkSession


def main():
    spark = (
        SparkSession.builder
```

```
        .appName("FirstPySparkProgram")
        .master("local[*]")
        .getOrCreate()
    )

    try:
        data = [
            (1, "Alice"),
            (2, "Bob"),
            (3, "Charlie"),
        ]

        df = spark.createDataFrame(data, ["id", "name"])

        print("PySpark is working! Here is the DataFrame:")
        df.show()
    finally:
        spark.stop()


if __name__ == "__main__":
    main()
```

5. Press **Ctrl + S** to save.

What the code does, in plain words: it starts Spark on your own computer (`local[*]` means "use all my CPU cores"), builds a small table with 3 people, prints it, and then stops Spark.

7.2 Create `requirements.txt`

This file lists the package your project needs.

1. Click the **New File** icon again.
2. Type `requirements.txt` and press **Enter**.
3. Type this single line:

```
pyspark==4.3.0
```

4. Press **Ctrl + S**.

To reinstall later, activate `.venv` and run `python -m pip install -r requirements.txt`. Java still needs its own install.

7.3 Create `.gitignore` (optional)

This file keeps local clutter out of Git, if you use Git later.

1. Click **New File**, type `.gitignore`, press **Enter**.
2. Paste this:

```
.venv/
venv/
__pycache__/
*.py[cod]
.vscode/
spark-warehouse/
metastore_db/
derby.log
.env
```

3. Press **Ctrl + S**.

Never put passwords or secret keys in files you plan to share.

7.4 Check the files exist

In the terminal, run:

```
dir
```

You should see `first_spark.py`, `requirements.txt` and `.gitignore`. If a name ends in `.txt` by mistake (for example `first_spark.py.txt`), right-click it in the Explorer, choose **Rename**, and fix the name.

Step 8: Run your first Spark program

You can run it in two ways. Pick one.

Way A: Run from the terminal

Check that `(.venv)` shows at the start of the terminal line, then run:

```
python first_spark.py
```

Way B: Use the Run button

1. Click the `first_spark.py` tab in the editor.
2. Click the ▷ **Run Python File** button at the top-right of the editor.

What you should see

The first run is slow because Spark is starting. Some extra log lines are normal unless an error stops the program. At the end you should see a table like this (spacing may differ slightly):

```
+---+-------+
| id|   name|
+---+-------+
|  1|  Alice|
|  2|    Bob|
|  3|Charlie|
+---+-------+
```

If you see the table with Alice, Bob and Charlie, **your local PySpark setup works.**

Coming back later

When you reopen VS Code another day:

1. Click **File → Open Folder** and choose `pyspark-basic`.
2. Open **Terminal → New Terminal** and choose **Command Prompt**.
3. Run `.venv\Scripts\activate.bat`.
4. Run `python first_spark.py`.

If something goes wrong

Find your message in the left column and do what the right column says.

Message or problem	What it means	Fix
`py` is not recognized	Python is not installed, or VS Code was not restarted	Redo Step 2, then close and reopen VS Code

Message or problem	What it means	Fix
`winget` is not recognized	Your Windows has no winget	Use the manual download links in Step 2 and Step 3
`java` is not recognized	Java is missing, or the terminal is old	Redo Step 3, then close and reopen VS Code
`JAVA_HOME` prints `%JAVA_HOME%`	The variable is not set for this terminal	Do Step 3.3, then reopen VS Code
No `(.venv)` at the start of the line	The environment is not switched on	Run `.venv\Scripts\activate.bat` from the `pyspark-basic` folder
`No module named pyspark`	PySpark is not in this environment	Activate `.venv`, then run `python -m pip install pyspark==4.3.0`
`first_spark.py` cannot be found	The terminal is in the wrong folder	Run `cd` and `dir`; go back with `cd /d "%USERPROFILE%\pyspark-basic"`
Script is not allowed to run (red text)	You are in PowerShell	Switch the terminal to **Command Prompt** (Step 1.3)
Spark fails to start	Java is missing or wrong	Run `java -version` and `echo %JAVA_HOME%`; you need Java 17+ and a `JAVA_HOME` pointing to the JDK main folder

Still stuck? Copy the **last red error line** and search for it, or send it to me and I will help.

Final folder and links

In the VS Code Explorer, your project should look like this:

```
pyspark-basic\
├── .venv\              (created automatically, do not edit)
├── first_spark.py
├── requirements.txt
└── .gitignore
```

Official pages used in this guide:

- VS Code download
- Python for Windows

- Eclipse Temurin JDK downloads
- PySpark installation documentation

Congratulations! You installed Python, Java and PySpark, and ran a Spark program, all from VS Code.